Homework 2

Big Data (BAX 423)

Instructor: Rahul Makhijani

Due: April 17, 2026 - 11:59 pm

On Canvas you will submit the following files in a single zip folder:

- A write-up (pdf/word doc) of your responses to Part 1.
- For questions that require a written response in Parts 2, 3 and 4 a pdf or word doc of your responses.
- iPython notebooks with all cells ran for Parts 2, 3 and 4 and the JSON files your notebooks generated.
- A short video (.mp4) of you describing what your group did in Part 3 for the multiclass classification problem.

Only one person should submit for your group. You may, but don't have to, submit your written responses to Parts 1-4 in a single document.

Part 1 - ML Design

In this section you will be considering the following production use case of Bloom Filters. Submit your responses in a pdf or word file.

NB: You may find it helpful to make us of the BloomFilterDemo notebook available on canvas in the Lecture 2 folder.

StreamEdge operates a global CDN with 500 edge servers across five regions. Each edge server caches up to **n = 2,000,000** video segments and uses a Bloom filter held entirely in RAM to avoid costly on-disk cache-index lookups on every request. The current deployment uses a bit-array of size **m = 19,170,117** and **k = 6** independent hash functions. As the platform grows, the engineering team must reason carefully about false positive rates, memory budgets, and the limits of standard Bloom filter design.

a. Using the standard approximation formula, calculate the false positive probability p for StreamEdge's configuration when the filter is filled to capacity (n = 2,000,000 items). Show your full work. Express your final answer as a percentage rounded to two decimal places.

b. The team wants to halve the false positive rate computed in Question 1 without changing n. Describe which parameter(s) they should adjust and in which direction. Then explain the trade-off this introduces in terms of memory usage, and calculate the new value of m required to achieve this halved rate (keeping k = 6).

c. StreamEdge's cache evicts stale segments daily. A junior engineer suggests simply deleting the corresponding bits in the Bloom filter when a segment is evicted. Explain why this approach is incorrect, and describe (in brief) one data structure that extends the Bloom filter to support deletion.

d. StreamEdge wants a Bloom filter that adapts as items are added — starting small and expanding without rebuilding from scratch. Describe the architecture of a Scalable Bloom Filter (SBF), and explain how it maintains a maximum false positive guarantee across multiple internal filters.

Part 2 - Anomaly Detection with Countmin Sketch

In this section we will be using the Intrusion Detection Evaluation Dataset (CIC-IDS2017), a file of source-to-destination web traffic flows and testing how reliably Countmin Sketch can be used to identify DoS (Denial of Service) attacks.

a. Download the dataset Wednesday-workingHours.pcap_ISCX.csv on Canvas under HW2/Datafiles.

b. Download the notebook CMS from Canvas.

c. Complete all TODOs in the notebook.

d. Submit your notebook with the output of all cells visible and the JSON file it outputs.

e. Submit your responses to written questions in a pdf or word file.

Part 3 - Anomaly Detection with LightGBM

In this section, we will be using the UNSW-NB15 Dataset, a benchmark dataset for web anomaly detection. Here rather than using merely the volume of traffic we will be trying to identify attacks with a high degree of accuracy and speed using other features of the requests and their origins.

a. Download the dataset UNSW-NB15_1.csv on Canvas under HW2/Datafiles. You will also need to download the feature names from the file UNSW-NB15_features.csv

b. Download the notebook UNSW_LightGBM.ipynb from Canvas.

c. Complete all TODOs in the notebook.

d. Submit your notebook with the output of all cells visible and the JSON file it outputs.

e. Submit your responses to written questions in a pdf or word file.

Part 4 - Range Queries

In this section we will be revisiting Countmin Sketch and constructing *range queries*. That is, we want to efficiently estimate counts between arbitrary intervals in the data. We will be using this to make simple approximations of text similarity.

a. Download Frankenstein.txt from Canvas under HW2/Datafiles .

b. Download the notebook RangeQueryText.ipynb on Canvas.

c. Complete all TODOs in the notebook.

d. Submit your notebook with the output of all cells visible and the JSON file it outputs.

e. Submit your responses to written questions in a pdf or word file.